Lecture 11: Review

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Review

Introduction

Outline

- Big Pictures
- Linearity
- No Perfect Collinearity
- Exogeneity (Conditional Mean)
- Conditional Variance and Covariance
- Distribution
- Unit Roots and Cointegration
- Forecasting, VAR, and Local Projections

Review

Big Pictures

Big Picture of Econometric Analysis

- Econometric analysis proceeds in five steps:
  1. **Definition**
  2. **Data**
  3. **Identification**
  4. **Estimation**
  5. **Inference and/or Forecasting**

Big Picture of Time-Series Models

Intertemporal

- Moving Average (MA) model of order q:

$$y_t = \gamma_0 + \gamma_1 \varepsilon_{t-1} + \ldots + \gamma_q \varepsilon_{t-q} + \varepsilon_t$$

- Autoregressive (AR) model of order p:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \ldots + \beta_p y_{t-p} + \varepsilon_t$$

- Autoregressive Moving Average (ARMA) model of order p and q:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \ldots + \beta_p y_{t-p} + \varepsilon_t + \gamma_1 \varepsilon_{t-1} + \ldots + \gamma_q \varepsilon_{t-q}$$

Big Picture of Time-Series Models

Intratemporal

- Static model:

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 z_t + \varepsilon_t$$

Big Picture of Time-Series Models

Intertemporal and intratemporal

- Finite distributed lag (FDL) model of order r:

$$y_t = \beta_0 + \beta_1 z_t + \beta_2 z_{t-1} + \ldots + \beta_{r+1} z_{t-r} + \varepsilon_t$$

- Autoregressive finite distributed lag (AR-FDL) model of order p and r :

$$y_t = \beta_0 + \beta_1 y_{t-1} + \ldots + \beta_p y_{t-p} + \gamma_1 z_t + \gamma_2 z_{t-1} + \ldots + \gamma_{r+1} z_{t-r} + \varepsilon_t$$

Big Picture of Time-Series Models

Time trend and cyclicality

- Time-series model with a time trend:

$$y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + \gamma t + \varepsilon_t$$

- Time-series model with seasonal components:

$$y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + \gamma_1 D_{Jan} + \gamma_1 D_{Feb} + \ldots + \gamma_{12} D_{Dec} + \varepsilon_t,$$

whhere D_* is a dammy variable indicating whether time t corresponds to month $*$.

Big Picture of This Course

	Topics	Goal
Lecture 1	Basic Concepts of Time-Series Analysis I	
Lecture 2	Basic Concepts of Time-Series Analysis II	**Mathmatical Properties**
Lecture 3	Basic Concepts of Time-Series Analysis III	
Lecture 4	Linear Regression: Finite-Sample Properties	
Lecture 5	Linear Regression: Large-Sample Properties I	**Estimatin and Inference**
Lecture 6	Linear Regression: Large-Sample Properties II	
Lecture 7	Unit Roots and Cointegration	
Lecture 8	Forecasting	
Lecture 9	VAR Models	**Forecasting**
Lecture 10	Local Projections	

Big Picture of This Course: Lectures 1–3

- We studied fundamental properties of time-series processes.
  - Time trend, seasonality, and stochastic components
  - Strong stationarity and weak stationarity
  - WN, IWN, and GWN
  - Random walk
  - MA, AR, and ARMA processes
  - Characteristic equation and ACF
  - Yule-Walker equation

Big Picture of This Course: Lectures 4–10

	Univariate				Multivariate	
	Finite-Sample	Large-Sample			VAR	LP
	Lecture 4	Lecture 5	Lecture 6	Lecture 7	Lecture 9	Lecture 10
Linearity	Stationarity or Non-stationarity	Weak stationarity &Weak dependence	Weak stationarity &Weak dependence	$\star$	Weak stationarity &Weak dependence	Weak stationarity &Weak dependence
Collinearity	No	No	No	No	No	No
Exogeneity	Strict	Contemporaneous	Contemporaneous	Contemporaneous	Contemporaneous	Contemporaneous
Variance	Homoskedasticity	Homoskedasticity	Heteroskedasticity	$*$	$*$	Heteroskedasticity
Covariance	Zero	Zero	Non-zero	$*$	$*$	Non-zero
Distribution	Normal	—	—	—	—	—

- $\star$: Unit roots and cointegration
- $*$: case-by-case
- Lecture 8: Forecasting

Big Picture of This Course: Key Takeaways

Goal of this course

- To be able to pick up the right framework for your analysis.

  $\hookrightarrow$ The balance between the model- and data-side structures.

Technical lesson

- Time-series econometrics is an application of the canonical linear regression analysis.

Broad lesson

- Knowledge is built from the bottom up.

Review

Linearity

Linearity

Assumption 1 (TS1): Linearity

The stochastic process $\{(y_t, x_t)\}_{t=0}^{T}$ follows

$$y_t = \beta_1 x_{t1} + \beta_2 x_{t2} + \ldots + \beta_K x_{tK} + \varepsilon_t,$$

where $\{\varepsilon_t\}_{t=0}^{T}$ is the sequence of unobserved errors.

- $\{y_t\}$, $\{x_t\}$, and $\{\varepsilon_t\}$: No stationarity-requirement.

Linearity

Assumption 1 (TS1'): Linearity

A jointly weakly stationary and weakly dependent stochastic process $\{(y_t, x_t)\}_{t=0}^{T}$ follows

$$y_t = \beta_1 x_{t1} + \beta_2 x_{t2} + \ldots + \beta_K x_{tK} + \varepsilon_t,$$

where $\{\varepsilon_t\}_{t=0}^{T}$ is a sequence of random errors.

- $\{y_t\}$ and $\{x_t\}$: weakly stationarity and weakly dependent.
- $\{\varepsilon_t\}$: no stationarity-requirement.
- $T \to \infty$

Review

Collinearity

Collinearity

Assumption 2 (TS2): No Perfect Collinearity

None of the explanatory variables is a perfect linear combination of the others.

- Intuition: There are no redundant regressors.
- This assumption is common in both finite- and large-sample theory.
- cf) Regularization (Ridge, LASSO, etc)

Review

Exogeneity (Conditional Expectation)

Exogeneity

Assumption 3 (TS3): Strict Exogeneity

$E[\varepsilon_t | X] = 0$ for all $t \in \{1, \ldots, T\}$.

Assumption 3 (TS3'): Contemporaneous Exogeneity

$E[\varepsilon_t | x_t] = 0$ for all $t \in \{1, \ldots, T\}$.

- Three configurations of exogeneity-type assumptions:
  - $s = t$: ε_t is uncorrelated with contemporaneous x_{sk}.
  - $s < t$: ε_t is uncorrelated with past x_{sk}.
  - $s > t$: ε_t is uncorrelated with future x_{sk}.

Review

Conditional Variance and Covariance

Conditional Variance and Covariance

Assumption 4-1 (TS4'-1): Homoskedasticity

$E[\varepsilon_t^2|x_t] = \sigma^2$ for all $t \in \{1, \ldots, T\}$.

Assumption 4-2 (TS4'-2): No Serial Correlation

$E[\varepsilon_t \varepsilon_s|x_t, x_s] = 0$ for all $t, s \in \{1, \ldots, T\}$ such that $t \neq s$.

$$E[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'] = \begin{bmatrix} \sigma^2 & 0 & \ldots & 0 \\ 0 & \sigma^2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \sigma^2 \end{bmatrix} = \sigma^2 \mathbf{I}$$

Conditional Variance and Covariance

Assumption 4-1 (TS4"-1): Heteroskedasticity

$E[\varepsilon_t^2|x_t] = \sigma_t^2$ for all $t \in \{1, \ldots, T\}$

Assumption 4-2 (TS4"-2): Serial Correlation

$E[\varepsilon_t \varepsilon_s|x_t, x_s] \neq 0$ for all $t, s \in \{1, \ldots, T\}$

$$E[\varepsilon\varepsilon'] = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \ldots & \sigma_{1T} \\ \sigma_{21} & \sigma_2^2 & \ldots & \sigma_{2T} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{T1} & \sigma_{T2} & \ldots & \sigma_T^2 \end{bmatrix} \neq \sigma^2 \mathbf{I}$$

- Note: $\sigma_{ij} = \sigma_{ji}$ for all i, j.

Conditional Variance and Covariance

- Under TS1'–TS3',

$$\sqrt{T}(\widehat{\beta}-\beta) \xrightarrow{d} \mathcal{N}(0, AVar(\widehat{\beta})),$$

  where $AVar(\widehat{\beta}) := (E[x_t x_t'])^{-1}\Omega(E[x_t x_t'])^{-1}$ with.

$$\Omega = \lim_{T\to\infty}\frac{1}{T}\Big\{\sum_{t=1}^{T} E[\varepsilon_t^2 x_t x_t'] + \sum_{t=1}^{T}\sum_{s=1}^{T} E[\varepsilon_t\varepsilon_s x_t x_s']\Big\}.$$

- Homoskedasticity and No serial correlation:

$$\Omega = \sigma^2 E[x_t x_t']$$

- Heteroskedasticity and Serial correlation:

$$\Omega = E[\varepsilon_t^2 x_t x_t'] + \sum_{h\neq 0} Cov(\varepsilon_t x_t, \varepsilon_{t-h} x_{t-h})$$

Conditional Variance and Covariance

- Newey-West variance estimator (a.k.a. HAC estimator):

  $\hookrightarrow$ directly estiamte each covariance term.

- GLS

  $\hookrightarrow$ transform data so as to satisfy homoskedasticity and no serial correlation.

Review

Distribution

Distribution

Assumption 5 (TS5): Normality

$\varepsilon_t | X \sim \mathcal{N}(0, \sigma^2)$ i.i.d. for $t \in \{1, \ldots, T\}$.

- Finite-sample theory relies on the distributional assumption.
- Large-sample theory is free from the distributional assumption.
  - CLT

Review

Unit Roots and Cointegration

Unit Roots

- An AR(1) process is weakly dependent.
  $\iff$ An AR(1) process is weakly stationary.
  $\iff$ The root of the characteristic equation is larger than one.

Proposition: Dickey-Fuller (DF) Test (Dickey, 1976; Fuller, 1976)

The t-statistic for testing

$$H_0 : \rho = 1 \text{ vs. } H_1 : \rho < 1 \text{ for } y_t = \rho y_{t-1} + \varepsilon_t$$
$$\iff H_0 : \theta = 0 \text{ vs. } H_1 : \theta < 0 \text{ for } \Delta y_t = \theta y_{t-1} + \varepsilon_t$$

is given by

$$t := \frac{\widehat{\theta} - 0}{se(\widehat{\theta})} = \frac{\widehat{\rho} - 1}{se(\widehat{\rho} - 1)} \xrightarrow{d} DF,$$

where $se(\widehat{\theta}) := (s^2 \sum_{t=1}^{T} y_{t-1}^2)^{1/2}$ with $s^2 := \frac{1}{T-1} \sum_{t=1}^{T} \{y_t - \widehat{\rho} y_{t-1}\}^2$. DF means the Dickey-Fuller distribution.

Unit Roots

- Consider an AR(p) model without a drift or a time trend:

$$y_t = \rho_1 y_{t-1} + \rho_2 y_{t-2} + \ldots + \rho_p y_{t-p} + \varepsilon_t \text{ with } \varepsilon_t \sim IWN(0, \sigma_\varepsilon^2).$$

- This can be transformed to

$$\Delta y_t = \underbrace{(\rho_1 - 1)}_{=:\theta} y_{t-1} + \zeta_1 \Delta y_{t-1} + \zeta_2 \Delta y_{t-2} + \ldots + \zeta_{p-1} \Delta y_{t-(p-1)} + \varepsilon_t.$$

- The unit root test boils down to testing

$$H_0 : \rho_1 = 1 \text{ vs. } H_1 : \rho_1 < 1 \qquad \Longleftrightarrow \qquad H_0 : \theta = 0 \text{ vs. } H_1 : \theta < 0$$

- The t statistic converges to the DF distribution:

$$t := \frac{\widehat{\theta} - 0}{se(\widehat{\theta})} = \frac{\widehat{\rho}_1 - 1}{se(\widehat{\rho}_1 - 1)} \xrightarrow{d} DF$$

- This is called the Augmented Dickey-Fuller (ADF) test (Dickey and Fuller, 1979).

Unit Roots

- The critical values vary across the following three cases:

(a) An AR(p) process without a drift nor a time trend:

$$y_t = \rho_1 y_{t-1} + \rho_2 y_{t-2} + \ldots + \rho_p y_{t-p} + \varepsilon_t$$
$$\hookrightarrow \Delta y_t = \theta y_{t-1} + \zeta_1 \Delta y_{t-1} + \zeta_2 \Delta y_{t-2} + \ldots + \zeta_{p-1} \Delta y_{t-(p-1)} + \varepsilon_t$$

(b) An AR(p) process with a drift but without a time trend:

$$y_t = \alpha + \rho_1 y_{t-1} + \rho_2 y_{t-2} + \ldots + \rho_p y_{t-p} + \varepsilon_t$$
$$\hookrightarrow \Delta y_t = \alpha + \theta y_{t-1} + \zeta_1 \Delta y_{t-1} + \zeta_2 \Delta y_{t-2} + \ldots + \zeta_{p-1} \Delta y_{t-(p-1)} + \varepsilon_t$$

(c) An AR(p) process with a drift and a time trend:

$$y_t = \alpha + \gamma t + \rho_1 y_{t-1} + \rho_2 y_{t-2} + \ldots + \rho_p y_{t-p} + \varepsilon_t$$
$$\hookrightarrow \Delta y_t = \alpha + \gamma t + \theta y_{t-1} + \zeta_1 \Delta y_{t-1} + \zeta_2 \Delta y_{t-2} + \ldots + \zeta_{p-1} \Delta y_{t-(p-1)} + \varepsilon_t$$

Dickey-Fuller Test Critical Values: (a) An AR(1) process without a drift nor a time trend

Sample size (T)	Probability that the statistic is less than entry							
	0.01	0.025	0.05	0.10	0.90	0.95	0.975	0.99
25	-2.65	-2.26	-1.95	-1.60	0.92	1.33	1.70	2.15
50	-2.62	-2.25	-1.95	-1.61	0.91	1.31	1.66	2.08
100	-2.60	-2.24	-1.95	-1.61	0.90	1.29	1.64	2.04
250	-2.58	-2.24	-1.95	-1.62	0.89	1.28	1.63	2.02
500	-2.58	-2.23	-1.95	-1.62	0.89	1.28	1.62	2.01
∞	-2.58	-2.23	-1.95	-1.62	0.89	1.28	1.62	2.01

Source: Dickey (1976), corrected in Fuller (1996).

Unit Roots

Dickey-Fuller Test Critical Values: (b) An AR(1) process with a drift but without a time trend

Sample size (T)	Probability that the statistic is less than entry							
	0.01	0.025	0.05	0.10	0.90	0.95	0.975	0.99
25	-3.75	-3.33	-2.99	-2.64	-0.37	0.00	0.34	0.71
50	-3.59	-3.23	-2.93	-2.60	-0.41	-0.04	0.28	0.66
100	-3.50	-3.17	-2.90	-2.59	-0.42	-0.06	0.26	0.63
250	-3.45	-3.14	-2.88	-2.58	-0.42	-0.07	0.24	0.62
500	-3.44	-3.13	-2.87	-2.57	-0.44	-0.07	0.24	0.61
∞	-3.42	-3.12	-2.86	-2.57	-0.44	-0.08	0.23	0.60

Source: Dickey (1976), corrected in Fuller (1996).

Dickey-Fuller Test Critical Values: (c) An AR(1) process with a drift and a time trend

Sample size (T)	Probability that the statistic is less than entry							
	0.01	0.025	0.05	0.10	0.90	0.95	0.975	0.99
25	-4.38	-3.95	-3.60	-3.24	-1.14	-0.81	-0.50	-0.15
50	-4.16	-3.80	-3.50	-3.18	-1.19	-0.87	-0.58	-0.24
100	-4.05	-3.73	-3.45	-3.15	-1.22	-0.90	-0.62	-0.28
250	-3.98	-3.69	-3.42	-3.13	-1.23	-0.92	-0.64	-0.31
500	-3.97	-3.67	-3.42	-3.13	-1.24	-0.93	-0.65	-0.32
∞	-3.96	-3.67	-3.41	-3.13	-1.25	-0.94	-0.66	-0.32

Source: Dickey (1976), corrected in Fuller (1996).

Unit Roots

- Ex) An AR(p) process with a drift and a time trend:

$$
\begin{aligned}
&y_t = \alpha + \gamma t + \rho_1 y_{t-1} + \rho_2 y_{t-2} + \ldots + \rho_p y_{t-p} + \varepsilon_t \\
&\hookrightarrow \Delta y_t = \alpha + \gamma t + \theta y_{t-1} + \zeta_1 \Delta y_{t-1} + \zeta_2 \Delta y_{t-2} + \ldots + \zeta_{p-1} \Delta y_{t-(p-1)} + \varepsilon_t.
\end{aligned}
$$

- The AIC suggests $p^* = 2 \longrightarrow$ effective sample size $= 203$
- The estimated model:

$$
\begin{aligned}
\Delta y_t = &\underset{(0.10)}{0.22} + \underset{(0.00011)}{0.00022}\ t - \underset{(0.0137)}{0.0293}\ y_{t-1} + \underset{(0.066)}{0.348}\ \Delta y_{t-1}.
\end{aligned}
$$

- The ADF t-statistic is

$$
t = \frac{-0.0293 - 0}{0.0137} \approx -2.13
$$

- The 5% critical value: $t_\alpha = -3.45 \sim -3.42$
- Since $t > t_\alpha$, do not reject H_0 at 5% significance level.

Cointegration

Definition: Integrated Processes of Order d

(i) A time-series process $\{y_t\}$ is an integrated process of order 0, denoted as $I(0)$, if it is weakly stationary.
(ii) A time-series process $\{y_t\}$ is an integrated process of order d, denoted as $I(d)$, if its dth difference is $I(0)$ (i.e., weakly stationary).

- The DF and ADF tests can be viewed as testing $I(0)$ vs. $I(1)$:

$$\Delta y_t = \theta y_{t-1} + \varepsilon_t,$$

with

$$\begin{cases} H_0 : \theta = 0 \\ H_1 : \theta < 0 \end{cases} \iff \begin{cases} H_0 : \{y_t\} \sim I(1) \\ H_1 : \{y_t\} \sim I(0) \end{cases}$$

Cointegration

Definition: Cointegration

Two processes $\{y_t\} \sim I(1)$ and $\{x_t\} \sim I(1)$ are said to be cointegrated if there exists a vector of constants $\gamma \in \mathbb{R}^K$ such that $\{y_t - \gamma' x_t\} \sim I(0)$.

- The existence of γ is often rationalized by the underlying theories:
  - The long-run equilibrium relationship between consumption and investment.
  - The abtrage relationship between financial assets.
- Cointegration test boils down to applying the DF or ADF test to $\{y_t - \gamma' x_t\}$.

Cointegration

- There are two scenarios depending on whether the cointegrating vector γ is known or not.
- **If γ is known:**

  (Step 1) Define the linear combination with the given values of γ:

  $$z_t := y_t - \gamma' x_t.$$

  (Step 2) Apply the DF (or ADF) test for z_t to study $H_0 : z_t \sim I(1)$ vs. $H_1 : z_t \sim I(0)$.

  - If H_0 is rejected, then $\{y_t\}$ and $\{x_t\}$ are cointegrated.

Cointegration

- **If γ is unknown:**

  (Step 1) Estimate γ by OLS, yeilding $\widehat{\gamma}$.

  (Step 2) Define the linear combination with the estimate $\widehat{\gamma}$:

$$z_t := y_t - \widehat{\gamma}' x_t.$$

  (Step 3) Apply the DF (or ADF) test for z_t to study $H_0 : z_t \sim I(1)$ vs. $H_1 : z_t \sim I(0)$.

  - If H_0 is rejected, then $\{y_t\}$ and $\{x_t\}$ are cointegrated.

- The OLS estimate $\widehat{\gamma}$ is calculated under H_0 (i.e., a spurious regression).

  $\Longrightarrow$ This issue has to be taken into account in the subsequent DF (or ADF) test.

Cointegration

m	Probability that the statistic is less than entry			
	0.01	0.025	0.05	0.10
Regression with a drift but without a time trend				
1	-3.90	-3.59	-3.34	-3.04
2	-4.29	-4.00	-3.74	-3.45
3	-4.64	-4.35	-4.10	-3.81
Regression with a drift and a time trend				
1	-4.32	-4.03	-3.78	-3.50
2	-4.66	-4.37	-4.12	-3.84
3	-4.97	-4.68	-4.43	-4.15

Source: Engle and Granger (1987) and Phillips and Ouliaris (1990), corrected in Davidson and MacKinnon (1993).

Cointegration

- If (and only if!) the cointegration test rejects H_0, linear regression of y_t onto x_t is meaningful.
- Q. How do we estimate the cointegrating vector?

  A. Two representation theorems:

  1. Dynamic Ordinary Least Squares (DOLS; Stock and Watson, 1993)

     $\hookrightarrow$ long-run relationship
  2. Error Correction Model (ECM; Engle and Granger, 1987)

     $\hookrightarrow$ short-run relationship
- Estimating the cointegrating vector is quite different from testing for cointegration!

Cointegration

- In practice, the econometric time-series analysis proceeds in multiple steps:
  1. Check if the time-series data are weakly stationary and weakly dependent.
     $\hookrightarrow$ Unit-root test

  2-a. If these two properties are satisfied, we can apply linear regression.

  2-b. If these two properties are not satisfied, we look into the details.
     $\hookrightarrow$ Cointegration test

  3-a. If there is a cointegration, we can estimate the long- and short-run relationships.
     $\hookrightarrow$ DOLS, ECM

  3-b. If there is no cointegration, we need to do more work...
     $\hookrightarrow$ Multi-cointegration, revise theory, and transform data, etc.

Review

Forecasting, VAR, and Local Projections

Forecasting

- The optimal h-step-ahead forecast:

$$y_{t+h|t} = E[y_{t+h}|\mathcal{I}_t],$$

  where $\mathcal{I}_t$ is the information set at (the end of) time t.

- To link y_{t+h} and $\mathcal{I}_t$, we need a model:
  - e.g., $y_{t+1} = \rho_0 + \rho_1 y_{t-1} + \varepsilon_{t+1}$.

- Two ways of forecasting:
  1. Point forecast $\widehat{y}_{t+h|t}$
  2. Interval forecast $CI_\alpha(\widehat{y}_{t+h|t})$, where α is the level of significance:
     - The sampling variation of $\widehat{y}_{t+h|t}$.
     - As-yet-realized future fluctuation in y_{t+h}.

Forecasting

- Impulse response function:

$$
\begin{aligned}
IRF_t(h) &:= \frac{\partial E[y_{t+h}|I_t]}{\partial \varepsilon_t} \\
&= \lim_{\alpha \to 0} \frac{E[y_{t+h}|I_{t-1} \cup \{\varepsilon_t + \alpha\}] - E[y_{t+h}|I_{t-1} \cup \{\varepsilon_t\}]}{\alpha}.
\end{aligned}
$$

i.e., the responsiveness of $y_{t+h|t}$ with respect to a marginal change in ε_t.

Forecasting

- Cumulative impulse response function:

$$CIRF_t(h) := \sum_{\ell=1}^{h} IRF_t(\ell)$$

  i.e., the accumulated effect of the change in ε_t from time t to $t+h$.

- Long-run impulse response function:

$$LRIRF_t(h) := \lim_{h\to\infty} CIRF_t(h)$$

  i.e., the full effect of the change in ε_t over all time.

VAR

- In constructing $IRF_t(h)$, we can incorporate the interactions between multiple variables.
- Structural Vector Autoregression Model (SVAR):

$$x_t = \alpha_0 + \alpha_1 z_t + \alpha_2 x_{t-1} + \alpha_3 z_{t-1} + \varepsilon_t^x$$
$$z_t = \beta_0 + \beta_1 x_t + \beta_2 x_{t-1} + \beta_3 z_{t-1} + \varepsilon_t^x,$$

  where $\varepsilon_t^x \sim WN(0, \sigma^2_{\varepsilon^x})$ and $\varepsilon_t^z \sim WN(0, \sigma^2_{\varepsilon^z})$ with $Cov(\varepsilon_t^x, \varepsilon_t^z) = 0$.
- α_1, β_1: effects of a unit change in contemporaneous variables.
- α_2, α_3, β_2, β_3,: effects of a unit change in lagged variables.

VAR

- In matrix notation,

$$A_0 Y_t = A_1 + A_2 Y_{t-1} + \boldsymbol{\varepsilon}_t,$$

where

$$Y_t := \begin{bmatrix} x_t \\ z_t \end{bmatrix} \quad \text{and} \quad \boldsymbol{\varepsilon}_t := \begin{bmatrix} \varepsilon_t^x \\ \varepsilon_t^z \end{bmatrix},$$

with

$$A_0 := \begin{bmatrix} 1 & -\alpha_1 \\ -\beta_1 & 1 \end{bmatrix}, \quad A_1 := \begin{bmatrix} \alpha_0 \\ \beta_0 \end{bmatrix}, \quad A_2 := \begin{bmatrix} \alpha_2 & \alpha_3 \\ \beta_2 & \beta_3 \end{bmatrix}, \quad \text{and} \quad \Xi_{\boldsymbol{\varepsilon}} := Var(\boldsymbol{\varepsilon}_t) = \begin{bmatrix} \sigma^2_{\varepsilon^x} & 0 \\ 0 & \sigma^2_{\varepsilon^z} \end{bmatrix}.$$

- A_0, A_1, A_2, and $\Xi_{\boldsymbol{\varepsilon}}$: structural parameters

VAR

- Reduce-form Vector Autoregressive Model:

$$Y_t = B_0 + B_1 Y_{t-1} + \boldsymbol{\eta}_t,$$

where

$$B_0 := A_0^{-1} A_1, \quad B_1 := A_0^{-1} A_2, \quad \text{and} \quad \boldsymbol{\eta}_t = \begin{bmatrix} \eta_t^x \\ \eta_t^z \end{bmatrix} := A_0^{-1} \boldsymbol{\varepsilon}_t.$$

with

$$\Xi_{\boldsymbol{\eta}} := Var(\boldsymbol{\eta}_t) = A_0^{-1} \Xi_{\boldsymbol{\varepsilon}} (A_0^{-1})'.$$

- B_0, B_1, and $\Xi_{\boldsymbol{\eta}}$: reduced-form parameters

VAR

- Under TS1'-TS3', the reduced-form parameters B_0, B_1, and $\Xi_{\boldsymbol{\eta}}$ are identified.

- In general, for n-variate VAR models:

$$\begin{cases} B_0 : n \text{ parameters} \\ B_1 : n^2 \text{ parameters} \\ \Xi_{\boldsymbol{\eta}} : \frac{n(n+1)}{2} \text{ parameters} \end{cases} \qquad \begin{cases} A_0 : n(n-1) \text{ parameters} \\ A_1 : n \text{ parameters} \\ A_2 : n^2 \text{ parameters} \\ \Xi_{\boldsymbol{\varepsilon}} : n \text{ parameters} \end{cases}$$

$\hookrightarrow \frac{3}{2}n(n+1)$ reduced-form parameters $\qquad \hookrightarrow 2n^2 + n$ structural parameters

- Identification of strutural paramters requres $\frac{1}{2}n(n-1)$ additional restrictions.
  - normalization, short-run restrictions, and long-run restrictions

Normalization

- The variance-covariance matrix:

$$\underbrace{\begin{bmatrix} \sigma^2_{\eta^x_t} & \sigma_{\eta^x_t \eta^z_t} \\ \sigma_{\eta^x_t \eta^z_t} & \sigma^2_{\eta^z_t} \end{bmatrix}}_{\Xi_{\boldsymbol{\eta}}} = \underbrace{\begin{bmatrix} 1 & -\alpha_1 \\ -\beta_1 & 1 \end{bmatrix}^{-1}}_{A_0^{-1}} \underbrace{\begin{bmatrix} \color{red}{\sigma^2_{\varepsilon^x_t}} & 0 \\ 0 & \sigma^2_{\varepsilon^z_t} \end{bmatrix}}_{\Xi_{\boldsymbol{\varepsilon}}} \underbrace{\begin{bmatrix} 1 & -\alpha_1 \\ -\beta_1 & 1 \end{bmatrix}^{-1\prime}}_{(A_0^{-1})'}$$

- By setting $\color{red}{\sigma^2_{\varepsilon^x_t} = 1}$,
  - HLS: 3 reduced-form parameters (i.e., $\sigma^2_{\eta^x_t}$, $\sigma^2_{\eta^z_t}$, and $\sigma_{\eta^x_t \eta^z_t}$)
  - RHS: 3 structural parameters (i.e., α_1, β_1, and $\sigma^2_{\varepsilon^z_t}$)

  $\Longrightarrow$ The structural parameter matrix A_0 is identified.

Short-run restriction

- The variance-covariance matrix:

$$\underbrace{\begin{bmatrix} \sigma^2_{\eta^x_t} & \sigma_{\eta^x_t \eta^z_t} \\ \sigma_{\eta^x_t \eta^z_t} & \sigma^2_{\eta^z_t} \end{bmatrix}}_{\Xi_{\boldsymbol{\eta}}} = \underbrace{\begin{bmatrix} 1 & -\alpha_1 \\ -\beta_1 & 1 \end{bmatrix}^{-1}}_{A_0^{-1}} \underbrace{\begin{bmatrix} \sigma^2_{\varepsilon^x_t} & 0 \\ 0 & \sigma^2_{\varepsilon^z_t} \end{bmatrix}}_{\Xi_{\boldsymbol{\varepsilon}}} \underbrace{\begin{bmatrix} 1 & -\alpha_1 \\ -\beta_1 & 1 \end{bmatrix}^{-1'}}_{(A_0^{-1})'}$$

- By setting $-\alpha_1 = 0$,
  - HLS: 3 reduced-form parameters (i.e., $\sigma^2_{\eta^x_t}$, $\sigma^2_{\eta^z_t}$, and $\sigma_{\eta^x_t \eta^z_t}$)
  - RHS: 3 structural parameters (i.e., β_1, $\sigma^2_{\varepsilon^x_t}$, and $\sigma^2_{\varepsilon^z_t}$)

  $\Longrightarrow$ The structural parameter matrix A_0 is identified.

VAR

- "A_0 is lower triangular" restricts the contemporaneous (or short-run) relationship:

  - in terms of variables:

$$\begin{bmatrix} x_t \\ z_t \end{bmatrix} = \begin{bmatrix} \alpha_0 \\ \beta_0 \end{bmatrix} + \begin{bmatrix} 0 & \color{red}{0} \\ \beta_1 & 0 \end{bmatrix} \begin{bmatrix} x_t \\ z_t \end{bmatrix} + \begin{bmatrix} \alpha_2 & \alpha_3 \\ \beta_2 & \beta_3 \end{bmatrix} \begin{bmatrix} x_{t-1} \\ z_{t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_t^x \\ \varepsilon_t^z \end{bmatrix}$$

  - in terms of shocks:

$$\begin{bmatrix} \eta_t^x \\ \eta_t^z \end{bmatrix} = \begin{bmatrix} 1 & \color{red}{0} \\ \beta_1 & 1 \end{bmatrix} \begin{bmatrix} \varepsilon_t^x \\ \varepsilon_t^z \end{bmatrix}$$

- The ordering is rationalized by institutional knowledge and timing assumptions.

Long-run restriction

- Economic theory may have implications for the long-run effects.
- The long-run impulse response function:

$$LRIRF_t := \lim_{h\to\infty} \sum_{\ell=1}^{h} IRF_t(\ell) = (I - B_1)^{-1} A_0^{-1}.$$

- Ex) Demand shocks ε_t^z only have a temporary impact on GDP growth x_t.
  $\Longrightarrow$ The (1,2) entry of $LRIRF_t$ has to be 0.

Local Projections

- $IRF_t(h) = B_1^h A_0$

  $\hookrightarrow$ Recovering B_1^h is the key to estimating $IRF_t(h)$.

- By repeated substitution,

$$
\begin{aligned}
Y_{t+h} &= \underbrace{\left(\sum_{j=0}^{h-1} B_1^j\right) B_0}_{C_0} + \underbrace{B_1^h}_{C_1} Y_t + \underbrace{\sum_{j=0}^{h} B_1^j \boldsymbol{\eta}_{t+h-j}}_{\boldsymbol{\nu}_t} \\
&= C_0 + C_1 Y_t + \boldsymbol{\nu}_t
\end{aligned}
$$

  $\hookrightarrow$ B_1^h can be recovered by directly regressing Y_{t+h} onto Y_t.

- This form is called the **direct regression** or **local projection** (Jordá, 2005).

Local Projections

- Simulation study by Li et al. (2024):

  - In many practically-relevant setups,

$$|Bias(\widehat{IRF}_t(h)^{LP})| < |Bias(\widehat{IRF}_t(h)^{VAR})|.$$

  - In view of the bias-variance tradeoff,

$$Var(\widehat{IRF}_t(h)^{LP}) > Var(\widehat{IRF}_t(h)^{VAR}).$$

- $\widehat{IRF}_t(h)^{LP}$ is more accurate, but less precise.
- $\widehat{IRF}_t(h)^{VAR}$ is less accurate, but more precise.

Local Projections

Large-sample properties:

- $\widehat{IRF}_t(h)^{LP}$ is consistent for $IRF_t(h)$ and asymptotically normally distributed.
  $\hookrightarrow$ $\widehat{IRF}_t(h)^{LP}$ is as accurate as $\widehat{IRF}_t(h)^{VAR}$.

Variance estiamtor:

- Recall:
$$Y_{t+h} = C'\tilde{X}_t + \boldsymbol{\nu}_t \quad \text{with} \quad \boldsymbol{\nu}_t := \sum_{j=0}^{h} B_1^j \boldsymbol{\eta}_{t+h-j} = \sum_{j=0}^{h} B_1^j A_0^{-1} \varepsilon_{t+h-j}$$

- $\{\boldsymbol{\nu}_t\}$ have serial correlation and heteroskedasticity.
  $\hookrightarrow$ The Newey-West (or HAC) estimator!

Summary and Preview

Summary

- Throughout this course, we have studied time-series econometrics.
- The choice of an appropriate framework depends on the model- and data-side structure.

Preview

- Assignment 3 will be uploaded by the end of today.
- Solution for Assignment 2 will be uploaded by the end of this week.
- Mock exam will be uploaded by the end of next week.
- Final exam: 8:30-10:30, June 12 (Fri), at Auditorium C - Hein Picard, Hoveniersberg, Campus Tweekerken

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